THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 2

Chapters 9–12: **Methods of Separation in Everyday Life, Living Creatures: Exploring their Characteristics, Nature's Treasures, Beyond Earth** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which is the correct pair? 1

(a) Venus: hottest planet **(b)** Mars: ringed
(c) Jupiter: smallest **(d)** Mercury: largest planet

2. When a hot saturated solution cools, the excess solute: 1

(a) crystallises out **(b)** evaporates **(c)** becomes gas **(d)** dissolves more

3. Frogs lay their eggs in: 1

(a) snow **(b)** water **(c)** sand **(d)** trees

4. Which of the following is recycled? 1

(a) air **(b)** ash **(c)** smoke **(d)** aluminium cans

5. A day on a planet is the time it takes to: 1

(a) cross **(b)** orbit
(c) revolve once **(d)** rotate once on its axis

6. Why is the filter paper folded and fitted in the funnel? 1

(a) to cool **(b)** to make it colourful
(c) to prevent filtration **(d)** to hold the paper well

7. A cow, a hen and a human are alike because all of them: 1

(a) cannot reproduce **(b)** need food and air
(c) live only in water **(d)** make their own food

8. In 200 L of air, the nitrogen is about: 1

(a) 21 L **(b)** 78 L **(c)** 42 L **(d)** 156 L

9. A lunar month is about: 1

(a) 1 day **(b)** 29.5 days **(c)** 365 days **(d)** 7 days

10. After the filtration above, the filtrate contains: 1

(a) chalk **(b)** sand **(c)** salt solution **(d)** iron

11. A robot dog moves, barks and responds to commands. Is it living? 1

(a) yes, it barks **(b)** yes, it moves
(c) no, it cannot grow, respire or reproduce **(d)** yes, it responds

12. Which gas in air does not support burning and makes up most of it? 1

(a) oxygen **(b)** nitrogen **(c)** hydrogen **(d)** helium

13. The reason Earth has liquid water is that: 1

(a) it has no air **(b)** it is farthest

(c) its temperature is suitable **(d)** it is closest to Sun

14. A student added 5 spoons of salt to 100 mL of water and 3 spoons dissolved. This means: 1

(a) all dissolved **(b)** water evaporated

(c) no salt dissolved **(d)** solution became saturated

15. Tadpoles breathe through: 1

(a) lungs **(b)** gills **(c)** nose **(d)** skin only

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do plants kept in the dark become pale? 2

17. Differentiate between reduce and reuse. 2

18. Why do astronauts float in a space station? 2

19. Why is the mud left on the filter paper called residue? 2

20. Why can a fish not live out of water? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain why the Moon is called a natural satellite. 3

22. Why can a sieve not separate chalk powder from water? 3

23. Why do you sweat when you exercise? 3

24. List some alternatives to fossil fuels. 3

OR

Explain the working of the telescope in observing stars. 3

25. Describe the experiment to prove that a salt solution has salt even if it looks clear. 3

OR

Give reasons: 3

(a) Seeds are living,

(b) Rock is non-living,

(c) Mushroom is living.

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Analyse how the separation methods used in cooking and farming are related. 5

OR

Differentiate between respiration in plants, fish and humans. 5

27. Explain how the demand for energy has increased and what we can do. 5

OR

Explain why the Earth is the only known planet with life. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND